

Bayesian learning

Bayes Theorem in ML

Bayes Theorem explains how to update the probability of a hypothesis when new evidence is observed. It combines prior knowledge with data to make better decisions under uncertainty and forms the basis of **Bayesian Inference** in machine learning.

- Handles uncertainty and noisy data effectively
- Supports probabilistic interpretation of model predictions
- Enables learning even with limited data
- Plays a key role in real-world applications such as risk analysis and diagnosis

Mathematical Formulation of Bayes Theorem

$$P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$$

Bayes Theorem describes the relationship between conditional probabilities

- **Posterior Probability** $P(A \mid B)$
 $P(A \mid B)$: the updated probability of hypothesis A after observing evidence B
- **Likelihood** $P(B \mid A)$
 $P(B \mid A)$: the probability of observing evidence B assuming hypothesis A is true.
- **Prior Probability** $P(A)$
 $P(A)$: the initial belief about hypothesis A before observing any evidence.
- **Evidence (Marginal Likelihood)** $P(B)$
 $P(B)$: the total probability of observing evidence B acting as a normalization factor.

Classification: Bayesian learning

- Bayes rule:
$$P(M | x) = \frac{P(x | M)P(M)}{P(x)} = \frac{P(x | M)P(M)}{\sum_M P(x | M)P(M)}$$
- To classify a given datapoint $\mathbf{X}$ we select the model (class) M_i with the highest $P(M_i | x)$
- The denominator is a normalizing term and does not affect the classification of a given datapoint. Therefore
$$P(M | x) \propto P(x | M)P(M)$$
- $P(x|M)$ is called the likelihood and $P(M)$ is the prior probability. To classify a given datapoint x we need to know the likelihood and the prior.
- If priors $P(M)$ are uniform (the same) then finding the model that maximizes $P(M | D)$ is the same as finding M that maximizes the likelihood $P(D | M)$.

Maximum likelihood

- We can classify by simply selecting the model M that has the highest $P(M/D)$ where D =data, M =model. Thus classification can also be framed as the problem of finding M that maximizes $P(M/D)$

- By Bayes rule:

$$P(M | D) = \frac{P(D | M)P(M)}{P(D)} = \frac{P(D | M)P(M)}{\sum_M P(D | M)P(M)}$$

Maximum likelihood

- Suppose we have k models to consider and each has the same probability. In other words we have a uniform prior distribution $P(M)=1/k$. Then

$$P(M | D) = P(D | M) \frac{1/k}{\sum_M P(D | M)P(M)} \propto P(D | M)$$

- In this case we can solve the classification problem by finding the model that maximizes $P(D | M)$. This is called the maximum likelihood optimization criterion.

Maximum likelihood

- Suppose we have n i.i.d. samples (x_i, y_i) drawn from M . The likelihood $P(D|M)$ is

$$\begin{aligned} P(D|M) &= P((x_1, y_1), \dots, (x_n, y_n) | M) = P(x_1, y_1 | M) \dots P(x_n, y_n | M) \\ &= \prod_{i=1}^n P(x_i, y_i | M) = \prod_{i=1}^n P(y_i | x_i, M) P(x_i) \end{aligned}$$

- Consequently the log likelihood is

$$-\log P(D|M) = -\sum_{i=1}^n \log P(y_i | x_i, M) - \sum_{i=1}^n P(x_i)$$

Maximum likelihood and empirical risk

- Maximizing the likelihood $P(D/M)$ is the same as maximizing $\log(P(D/M))$ which is the same as minimizing $-\log(P(D/M))$

- Set the loss function to

$$c(x_i, y_i, f(x_i)) = -\log(P(y_i | x_i, f))$$

- Now minimizing the empirical risk is the same as maximizing the likelihood (return to this later again)

Maximum likelihood example

- Consider a set of coin tosses produced by a coin with $P(H)=p$ ($P(T)=1-p$)
- We want to determine the probability $P(H)$ of the coin that produces k heads and $n-k$ tails?
- We are given some tosses (training data):
HTHHHTHHHTH.
- Solution:
 - Form the log likelihood
 - Differentiate w.r.t. p
 - Set the derivative to 0 and solve for p

Maximum likelihood example

- Likelihood is probability of data given model
- Data are the set of coin tosses and model is given by one parameter p
- $P(data|p) = p^k(1 - p)^{n-k}$
- $Log(P(data|p)) = \log(p^k) + \log(1 - p)^{n-k}$
 $= k \log(p) + (n - k)\log(1 - p)$
- Take derivative with respect to p , set to 0, and solve for p

Classification by likelihood

- Suppose we have two classes C_1 and C_2 .
- Compute the likelihoods $P(D|C_1)$ and $P(D|C_2)$.
- To classify test data D' assign it to class C_1 if $P(D|C_1)$ is greater than $P(D|C_2)$ and C_2 otherwise.

Gaussian models

- Assume that class likelihood is represented by a Gaussian distribution with parameters μ (mean) and σ (standard deviation)

$$P(x | C_1) = \frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}} \quad P(x | C_2) = \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(x-\mu_2)^2}{2\sigma_2^2}}$$

- We find the model (in other words mean and variance) that maximize the likelihood (or equivalently the log likelihood). Suppose we are given

training points $x_1, x_2, \dots, x_{n1}$ from class C_1 . Assuming that each datapoint is drawn independently from C_1 the sample log likelihood is

$$P(x_1, x_2, \dots, x_{n1} | C_1) = P(x_1 | C_1) P(x_2 | C_1) \dots P(x_{n1} | C_1) = \frac{1}{\sqrt[n1]{2\pi}\sigma_1} e^{-\frac{\sum_{i=1}^{n1} (x_i - \mu_1)^2}{2\sigma_1^2}}$$

Gaussian models

- The log likelihood is given by

$$\log(P(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{n_1} | \mathcal{C}_1)) = -\frac{n_1}{2} \log(2\pi) - n_1 \log(\sigma_1) - \frac{\sum_{i=1}^{n_1} (\mathbf{x}_i - \mu_1)^2}{2\sigma_1^2}$$

- By setting the first derivatives $dP/d\mu_1$ and $dP/d\sigma_1$ to 0. This gives us the maximum likelihood estimate of μ_1 and σ_1 (denoted as $\mathbf{m}_1$ and $\mathbf{s}_1$ respectively)

$$\mathbf{m}_1 = \frac{\sum_{i=1}^{n_1} \mathbf{x}_i}{n_1} \quad \mathbf{s}_1^2 = \frac{\sum_{i=1}^{n_1} (\mathbf{x}_i - \mathbf{m}_1)^2}{n_1}$$

- Similarly we determine $\mathbf{m}_2$ and $\mathbf{s}_2$ for class $\mathcal{C}_2$.

Gaussian models

- After having determined class parameters for C_1 and C_2 we can classify a given datapoint by evaluating $P(x|C_1)$ and $P(x|C_2)$ and assigning it to the class with the higher likelihood (or log likelihood).

$$\log(P(x | C_1)) = -\frac{1}{2}\log(2\pi) - \log(s_1) - \frac{(x_i - m_1)^2}{2s_1^2}$$

$$\log(P(x | C_2)) = -\frac{1}{2}\log(2\pi) - \log(s_2) - \frac{(x_i - m_2)^2}{2s_2^2}$$

- The likelihood can also be used as a loss function and has an equivalent representation in empirical risk minimization (return to this later).

Gaussian classification example

- Consider one dimensional data for two classes (SNP genotypes for case and control subjects).

- Case (class C_1): 1, 1, 2, 1, 0, 2

- Control (class C_2): 0, 1, 0, 0, 1, 1

- Under the Gaussian assumption case and control classes are represented by Gaussian distributions with parameters (μ_1, σ_1) and (μ_2, σ_2) respectively. The maximum likelihood estimates of means are

estimates of means are

$$m_1 = \frac{\sum_{i=1}^{n_1} x_i}{n_1} = \frac{1+1+2+1+0+2}{6} = 7/6$$

$$m_2 = \frac{0+1+0+0+1+1}{6} = 3/6$$

Gaussian classification example

- The estimates of class standard deviations are

$$s_1 = \frac{\sum_{i=1}^{n_1} (x_i - m_1)^2}{n_1} = \frac{(1-7/6)^2 + (1-7/6)^2 + (2-7/6)^2 + (1-7/6)^2 + (0-7/6)^2 + (2-7/6)^2}{6} = .47$$

- Similarly $s_2 = .25$
- Which class does $x=1$ belong to? What about $x=0$ and $x=2$?

$$\log(P(x | C_1)) = -\frac{1}{2} \log(2\pi) - \log(s_1) - \frac{(x_i - m_1)^2}{2s_1^2}$$

$$\log(P(x | C_2)) = -\frac{1}{2} \log(2\pi) - \log(s_2) - \frac{(x_i - m_2)^2}{2s_2^2}$$

- What happens if class variances are equal?

Multivariate Gaussian classification

- Suppose each datapoint is an m -dimensional vector. In the previous example we would have m SNP genotypes instead of one. The class likelihood is given by

$$P(x | C_1) = \frac{1}{(2\pi)^{d/2} |\Sigma_1|^{1/2}} e^{-\frac{1}{2}(x-\mu_1)^T \Sigma_1^{-1} (x-\mu_1)}$$

- Where Σ_1 is the class covariance matrix. Σ_1 is of dimension $d \times d$. The $(i,j)^{\text{th}}$ entry of Σ_1 is the covariance of the i^{th} and j^{th} variable.

Multivariate Gaussian classification

- The maximum likelihood estimates of η_1 and Σ_1 are

$$m_1 = \frac{\sum_{i=1}^{n_1} x_i}{n_1} \quad S_1 = \frac{\sum_{i=1}^{n_1} (x_i - m_1)(x_i - m_1)^T}{n_1}$$

- The class log likelihoods with estimated parameters (ignoring constant terms) are

$$\log(P(x | C_1)) = -\frac{1}{2} \log(|S_1|) - \frac{1}{2} (x - m_1)^T S_1^{-1} (x - m_1)$$

$$\log(P(x | C_2)) = -\frac{1}{2} \log(|S_2|) - \frac{1}{2} (x - m_2)^T S_2^{-1} (x - m_2)$$

Multivariate Gaussian classification

- If $S_1 = S_2$ then the class log likelihoods with estimated parameters (ignoring constant terms) are

$$\log(P(x | C_1)) = -\frac{1}{2}(x - m_1)^T S^{-1}(x - m_1)$$

- Depends on distance to means.

Naïve Bayes algorithm

- If we assume that variables are independent (no interaction between SNPs) then the off-diagonal terms of S are zero and the log likelihood becomes (ignoring constant terms)

$$\log(P(\mathbf{x} | C_1)) = -\frac{1}{2} \log\left(\prod_{j=1}^m s_{1j}\right) - \frac{1}{2} \sum_{j=1}^m \left(\frac{x_j - m_{1j}}{s_{1j}} \right)^2$$

Nearest means classifier

- If we assume all variances s_j to be equal then (ignoring constant terms) we get

$$\log(P(x | C_1)) = -\frac{1}{2s^2} \sum_{j=1}^m (x_j - m_{1j})^2$$

Gaussian classification example

- Consider three SNP genotype for case and control subjects.
 - Case (class C_1): (1,2,0), (2,2,0), (2,2,0), (2,1,1), (0,2,1), (2,1,0)
 - Control (class C_2): (0,1,2), (1,1,1), (1,0,2), (1,0,0), (0,0,2), (0,1,0)
- Classify (1,2,1) and (0,0,1) with the nearest means classifier